Sequences & Series

ClipSAT Printable Practice Suite (clipsat.org) • Topic 6.01

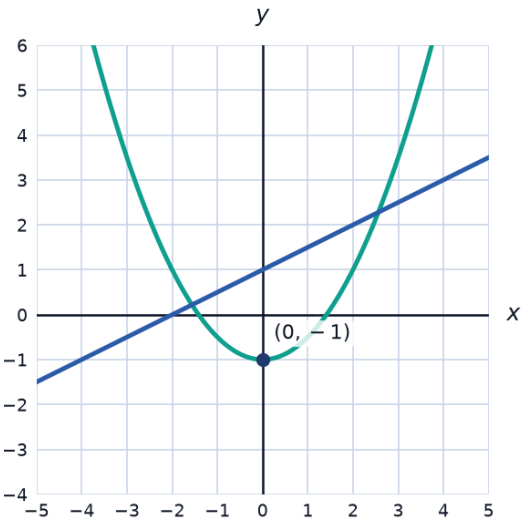
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: General Binomial Expansion $(1 + x)^n$ for Rational n

Find the coefficient of x^6 in the binomial expansion of $(1 + x)^{-3}$.

2 [Foundational] • Target: Approximating Roots via Binomial Series

Find the coefficient of x^3 in the binomial expansion of $(1 + x)^{-1}$.



3 [Foundational] • Target: Method of Differences & Telescoping Series

Find the coefficient of x^5 in the binomial expansion of $(1 + x)^5$.

4 [Foundational] • Target: Maclaurin Series & Higher-Order Approximations

Find the coefficient of x^2 in the binomial expansion of $(1 + x)^{-3}$.

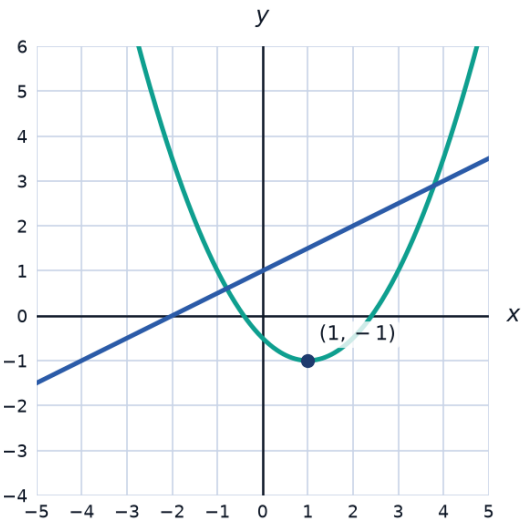
5 [Foundational] • Target: General Binomial Expansion $(1 + x)^n$ for Rational n

Find the coefficient of x^2 in the binomial expansion of $(1 + x)^4$.

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6 [Foundational] • Target: Approximating Roots via Binomial Series

Find the coefficient of x^4 in the binomial expansion of $(1 + x)^2$.



7 [Foundational] • Target: Method of Differences & Telescoping Series

The rational function f is defined by $f(x) = \frac{2x^4 - 3x^3 + kx^2 - 8x + 3}{(x - 2)(x^2 + 1)}$, where k is a real constant. Given that the series expansion of $f(x)$ in ascending powers of x, valid for sufficiently small $|x|$, has no term in x^2, find the value of k.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Maclaurin Series & Higher-Order Approximations

Find the coefficient of x^5 in the Maclaurin series of e^x.

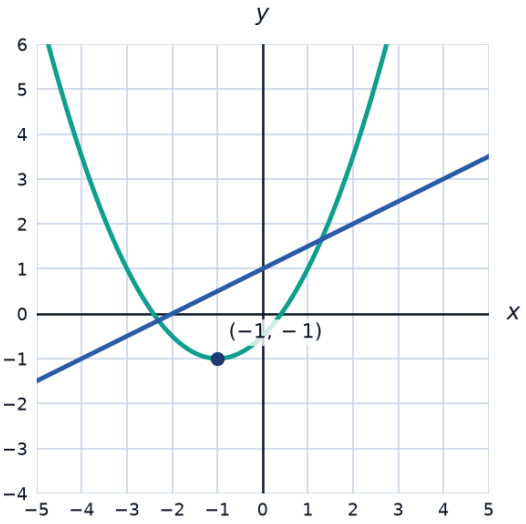
9 [Intermediate] • Target: General Binomial Expansion $(1 + x)^n$ for Rational n

Find the coefficient of x^6 in the Maclaurin series of e^x.

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10 [Intermediate] • Target: Approximating Roots via Binomial Series

Find the coefficient of x^4 in the Maclaurin series of e^x.



11 [Intermediate] • Target: Method of Differences & Telescoping Series

Find the coefficient of x^3 in the Maclaurin series of e^x.

12 [Intermediate] • Target: Maclaurin Series & Higher-Order Approximations

Find the coefficient of x^5 in the Maclaurin series of $\sin x$.

13 [Intermediate] • Target: General Binomial Expansion $(1 + x)^n$ for Rational n

Find the coefficient of x^6 in the Maclaurin series of $\cos x$.

14 [Intermediate] • Target: Approximating Roots via Binomial Series

Find the coefficient of x^4 in the Maclaurin series of $\cos x$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

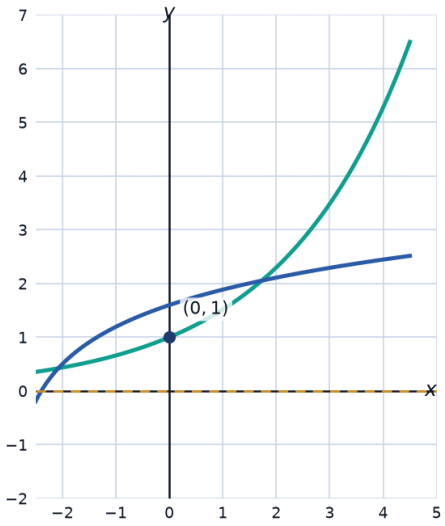
15 [Challenging] • Target: Method of Differences & Telescoping Series • Cross-Lesson: 7.01 Vectors

Find the coefficient of x^4 in the Maclaurin series of $\ln(1 + x)$.

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16 [Challenging] • Target: Maclaurin Series & Higher-Order Approximations • Cross-Lesson: 8.01 Algebra: Partial Fractions & Division

Find the coefficient of x^2 in the Maclaurin series of $\ln(1+x)$.



17 [Challenging] • Target: General Binomial Expansion $(1 + x)^n$ for Rational n • Cross-Lesson: 9.01 Complex Numbers

Find the coefficient of x^6 in the Maclaurin series of $\ln(1+x)$.

18 [Challenging] • Target: Approximating Roots via Binomial Series • Cross-Lesson: 1.01 Functions: Modulus & Inverse

Find the coefficient of x^5 in the Maclaurin series of $\ln(1+x)$.

19 [Challenging] • Target: Method of Differences & Telescoping Series • Cross-Lesson: 10.01 Numerical Methods

Find the coefficient of x^3 in the Maclaurin series of $\ln(1+x)$.

20 [Challenging] • Target: Maclaurin Series & Higher-Order Approximations • Cross-Lesson: 11.01 Differential Equations

Find the coefficient of x^3 in the Maclaurin series of $\cos x$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: General Binomial Expansion $(1 + x)^n$ for Rational n

Find the first three terms in the expansion of $(1+4x)^{1/2}$ in ascending powers of x.

a. $1 + 2x - 2x^2$

b. $1 + 2x + 2x^2$

c. $1 + 4x - 8x^2$

d. $1 + 2x - 4x^2$

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22 [Foundational] • Target: General Binomial Expansion $(1 + x)^n$ for Rational n

State the range of values of x for which the binomial expansion of $(1 + 4x)^{1/2}$ is valid.

a. $|x| < \frac{1}{4}$

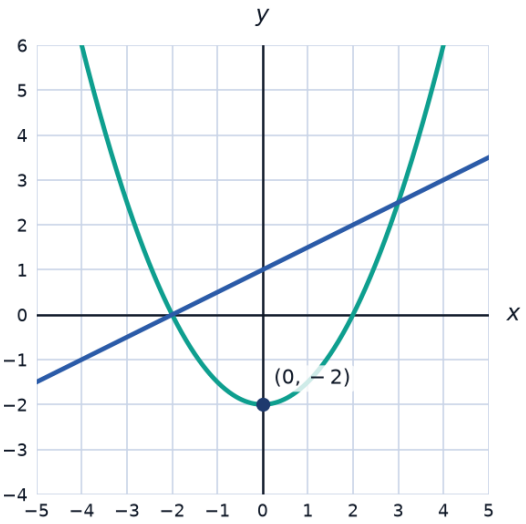
b. $|x| < 4$

c. $|x| < 1$

d. $x > 0$

23 [Foundational] • Target: General Binomial Expansion $(1 + x)^n$ for Rational n

Expand $(1 - 2x)^{-1}$ up to the term in x^3.



a. $1 + 2x + 4x^2 + 8x^3$

b. $1 - 2x + 4x^2 - 8x^3$

c. $1 + 2x + 2x^2 + 2x^3$

d. $1 - 2x - 4x^2 - 8x^3$

24 [Foundational] • Target: Approximating Roots via Binomial Series

Use the first three terms of $(1 - x)^{1/2} \approx 1 - \frac{1}{2}x - \frac{1}{8}x^2$ with $x = 0.02$ to approximate $\sqrt{0.98}$.

a. 0.98995

b. 0.98990

c. 0.99000

d. 0.98500

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25 [Foundational] • Target: Approximating Roots via Binomial Series

To approximate $\sqrt{50}$ using the binomial expansion of $(1+x)^{1/2}$, which factorization is most appropriate?

a. $\sqrt{49(1+1/49)} = 7(1+1/49)^{1/2}$

b. $\sqrt{100(1/2)} = 10(1/2)^{1/2}$

c. $(50)^{1/2} = (1+49)^{1/2}$

d. $\sqrt{36+14} = 6(1+14/36)^{1/2}$

26 [Foundational] • Target: Approximating Roots via Binomial Series

Find the coefficient of x^2 in the expansion of $\frac{1+2x}{(1-x)^2}$.

a. 5

b. 3

c. 4

d. 6

27 [Foundational] • Target: Method of Differences & Telescoping Series

Express $\frac{1}{r(r+1)}$ in partial fractions.

a. $\frac{1}{r} - \frac{1}{r+1}$

b. $\frac{1}{r} + \frac{1}{r+1}$

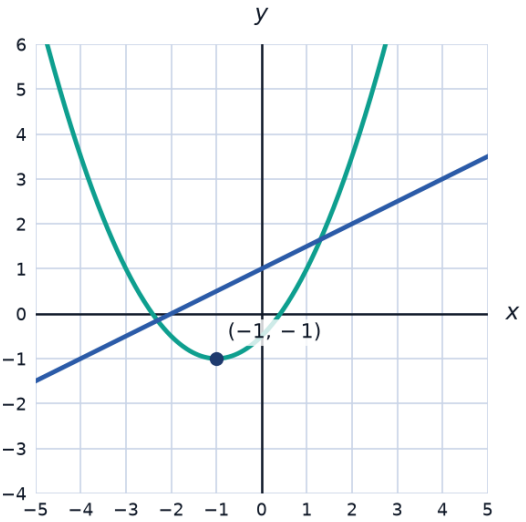
c. $\frac{1}{r+1} - \frac{1}{r}$

d. $\frac{2}{r} - \frac{1}{r+1}$

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28 [Foundational] • Target: Method of Differences & Telescoping Series

Use the method of differences to find $\sum_{r=1}^{n} \frac{1}{r(r+1)}$.



a. $\frac{n}{n+1}$

b. $1 - \frac{1}{n}$

c. $\frac{n+1}{n}$

d. $1 - \frac{1}{(n+1)^2}$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Method of Differences & Telescoping Series

Find the sum to infinity $\sum_{r=1}^{\infty} \frac{1}{r(r+1)}$.

a. 1

b. 2

c. 1/2

d. ∞

30 [Intermediate] • Target: Maclaurin Series & Higher-Order Approximations

State the general Maclaurin series formula for a function $f(x)$ expanded around $x = 0$.

a. $f(x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{3!}x^3 + \ldots$

b. $f(x) = f(0) + f'(0)x + f''(0)x^2 + f'''(0)x^3 + \ldots$

c. $f(x) = \sum_{n=0}^{\infty} f^{(n)}(x)\frac{x^n}{n!}$

d. $f(x) = f(1) + f'(1)(x-1) + \ldots$

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31 [Intermediate] • Target: Maclaurin Series & Higher-Order Approximations

Find the first three non-zero terms in the Maclaurin series of $\ln(1+x)$.

a. $x - \frac{x^2}{2} + \frac{x^3}{3}$

b. $x + \frac{x^2}{2} + \frac{x^3}{3}$

c. $x - x^2 + x^3$

d. $1 + x - \frac{x^2}{2}$

32 [Intermediate] • Target: Maclaurin Series & Higher-Order Approximations

Find the coefficient of x^3 in the Maclaurin series of $e^x \sin x$.

a. $\frac{1}{3}$

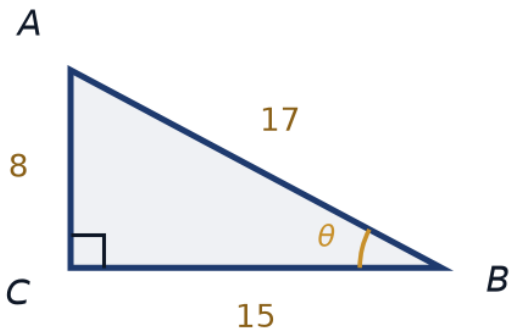
b. $\frac{1}{6}$

c. $-\frac{1}{3}$

d. 0

33 [Intermediate] • Target: General Binomial Expansion $(1 + x)^n$ for Rational n

Find the coefficient of x^1 in the Maclaurin series of $\sin x$.



a. -1

b. $\frac{13}{12}$

c. 1

d. $\frac{5}{6}$

34 [Intermediate] • Target: Approximating Roots via Binomial Series

Find the coefficient of x^5 in the Maclaurin series of $\cos x$.

a. $\frac{1}{120}$

b. 0

c. $-\frac{1}{120}$

d. $\frac{1}{5}$

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35 [Intermediate] • Target: Method of Differences & Telescoping Series

Find the coefficient of x^4 in the Maclaurin series of $\sin x$.

a. $\frac{1}{4}$

b. $-\frac{1}{24}$

c. 0

d. $\frac{1}{24}$

36 [Intermediate] • Target: Maclaurin Series & Higher-Order Approximations

The first three terms of a geometric progression are given by $u_1 = 3k - 2, \quad u_2 = k + 2, \quad u_3 = k - 2,$ where k is a real constant. Given that the infinite geometric series $\sum_{n=1}^{\infty} u_n$ converges, find the sum to infinity, S_∞.

a. 32

b. -1

c. $\frac{32}{3}$

d. 16

37 [Intermediate] • Target: General Binomial Expansion $(1 + x)^n$ for Rational n

The first three terms of a geometric progression are 32, 24, and 18. Find the sum to infinity, S_∞, of the progression.

a. 128

b. 96

c. 74

d. 64

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Approximating Roots via Binomial Series • Cross-Lesson: 1.01 Functions: Modulus & Inverse

The first term of a geometric progression is 16 and the fourth term is 2. Find the sum to infinity, S_∞, of the progression.

a. 32

b. $\frac{64}{3}$

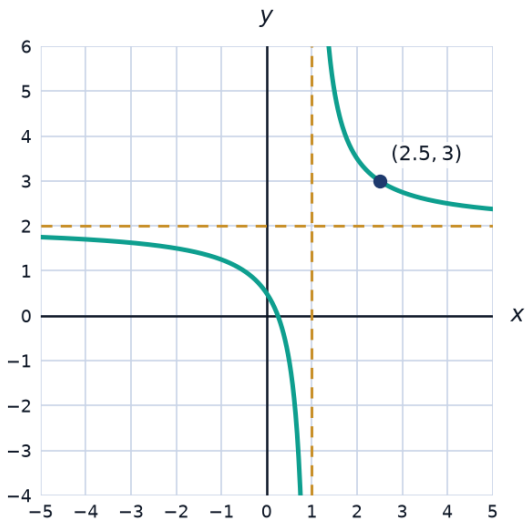
c. 24

d. 30

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39 [Challenging] • Target: Method of Differences & Telescoping Series • Cross-Lesson: 10.01 Numerical Methods

A sequence satisfies $u_{n+1} = \frac{3}{5}u_n + 7$ and converges to a limit L. Find L.



a. $\frac{49}{2}$

b. $\frac{35}{2}$

c. 7

d. $-\frac{35}{2}$

40 [Challenging] • Target: Maclaurin Series & Higher-Order Approximations • Cross-Lesson: 11.01 Differential Equations

A sequence satisfies $u_{n+1} = \frac{3}{5}u_n + 5$ and converges to a limit L. Find L.

a. $\frac{35}{2}$

b. $\frac{25}{2}$

c. 5

d. $-\frac{25}{2}$

41 [Challenging] • Target: General Binomial Expansion $(1 + x)^n$ for Rational n • Cross-Lesson: 12.01 Poisson Distribution

A sequence satisfies $u_{n+1} = \frac{1}{3}u_n + 1$ and converges to a limit L. Find L.

a. $\frac{5}{2}$

b. $-\frac{3}{2}$

c. 1

d. $\frac{3}{2}$

42 [Challenging] • Target: Approximating Roots via Binomial Series • Cross-Lesson: 13.01 Continuous Random Variables

A sequence satisfies $u_{n+1} = \frac{2}{5}u_n + 1$ and converges to a limit L. Find L.

a. $\frac{8}{3}$

b. $\frac{5}{3}$

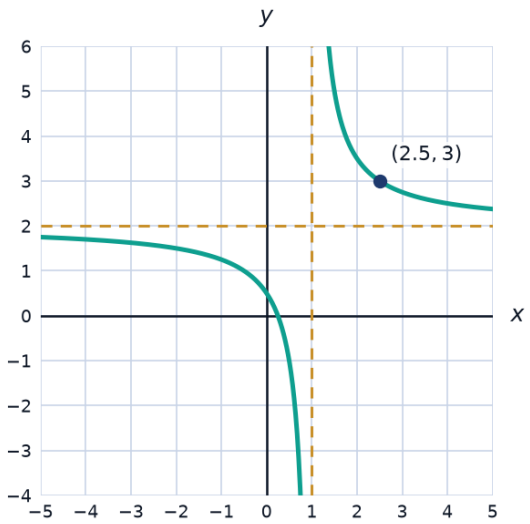
c. $-\frac{5}{3}$

d. 1

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43 [Challenging] • Target: Method of Differences & Telescoping Series • Cross-Lesson: 14.01 Sampling & Estimation

A sequence satisfies $u_{n+1} = \frac{1}{3}u_n + 3$ and converges to a limit L. Find L.



a. $\frac{15}{2}$

b. 3

c. $-\frac{9}{2}$

d. $\frac{9}{2}$

44 [Challenging] • Target: Maclaurin Series & Higher-Order Approximations • Cross-Lesson: 15.01 Hypothesis Testing

A sequence satisfies $u_{n+1} = \frac{1}{3}u_n + 8$ and converges to a limit L. Find L.

a. 12

b. -12

c. 20

d. 8

45 [Challenging] • Target: General Binomial Expansion $(1 + x)^n$ for Rational n • Cross-Lesson: 2.01 Exponentials & Logarithms

A geometric sequence has first term $a = 11$ and common ratio $r = 3$. Find the sum of the first 10 terms.

a. 324775

b. 216513

c. 324764

d. -324764

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